

Stub loaded compact hexagonal ring antenna for WLAN/WiMAX/ITU applications

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Abstract—A compact stub loaded hexagonal ring shape antenna using CPW feed is designed and implemented for WLAN/WiMAX/ITU application. Y- Shape stub is loaded inside hexagonal ring to achieve multiband performance with miniaturization. The proposed antenna is highly compact with $(20 \times 20 \times 0.8)$ mm³ dimension. The impedance bandwidth of the antenna is 3 GHz for both the band respectively from 2.9 to 5.9 GHz and 7 to 10 GHz. The substrate material used is Flame Retardant-4 (FR-4) due to its availability and low cost of fabrication. Relative dielectric constant and substrate thickness of material is 4.4 and 0.8 mm respectively. Ansys HFSS tool is used to optimize the designed parameters and experimental validation is done by Vector Network Analyzer.

Keywords—Y shape stub, CPW feed, multiband, hexagonal ring, miniaturization.

I. INTRODUCTION

Today's communication technology demands wide bandwidth to integrate various applications in a single device. Multiple wireless standards must be supported for versatility of the device. Due to its conformal nature, microstrip antennas are preferred. The low cost, simple fabrication and easy integration in different circuitry has made microstrip antennas very favorable. Currently different techniques of multiband and wideband antenna for WLAN/ WiMAX/ ITU/ X-Band have been proposed [1-4]. To achieve better performance in antenna application different type of slots and strips such as inverted F and L shaped slots used to create dual and triple band performance. Different shape of strip added in radiating element to achieve triple band characteristics [5, 6]. Ring type radiator and stub loading have unique properties which help to achieve both compactness and multiband response simultaneously [7]. Meta-material structure has negative permittivity and negative permeability which helps to realize multiband performance [8, 9]. Split Ring Resonator is a basic structure of permittivity; hence the improved performance can be achieved without affecting the size of radiating element. The Split Ring Resonator (SRR) and Complementary Split Ring Resonator (CSRR) based dual-

band and triple-band antenna were discussed [10, 11]. Fractal geometry also provides antennas for dual-band and wide-band applications [12]. Although mentioned methods are proposed for creating dual-band [1], triple-band [2, 3] and multi-band [4] structures with compact size, wide-band nature with multi-band application integrability in a single compact antenna was seldom discussed.

II. ANTENNA GEOMETRY

The proposed stub loaded antenna uses FR-4 substrate with relative dielectric constant and loss tangent of 4.4 and 0.02 respectively, with a 50Ω coplanar waveguide (CPW) transmission line feed. The thickness of the substrate is $X = 0.8$ mm. It has a feed length (F_L) of 7.5 mm and feed width F_W of 1.2 mm. Antennas of different evolution stages are shown in Fig.1 from #1 to #3.

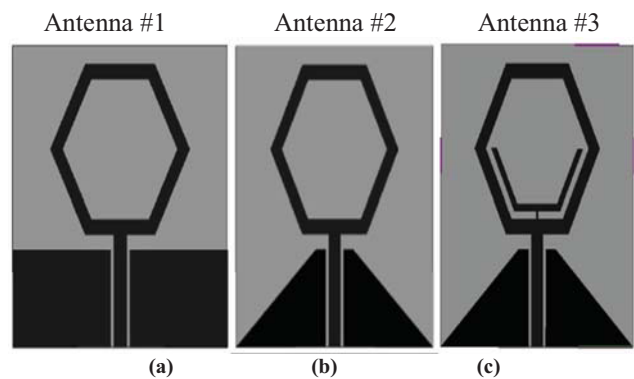


Fig.1: Design steps of antenna

Antenna #1 demonstrates an annular hexagonal ring antenna with a rectangular shaped ground plane where the hexagonal shaped ring is the primary radiating element. Antenna #2 exhibits tapering off of the ground plane. Hence improved impedance matching is obtained in higher frequency region at the cost of losing the resonant band at the lower frequency. Antenna #3 is the proposed stub loaded antenna, which includes the tapered ground idea while maintaining the previous dimensions ($L \times W$). The stub acts as a radiating element combined with the annular ring. The stub helps retain resonance at lower frequency while maintaining

higher resonance. The proposed antenna parameters are illustrated in Fig. 2. Dimensions of the final proposed prototype is presented in Table 1. The formulae for obtaining desired resonant bands has been shown Equations (1, 2 and 3). It should be noted that $k_{mn} = 1.84118$ for TM_{11} mode. Optimizations were made to the parameters to maximize the bandwidth of the dual-band antenna [14, 15].

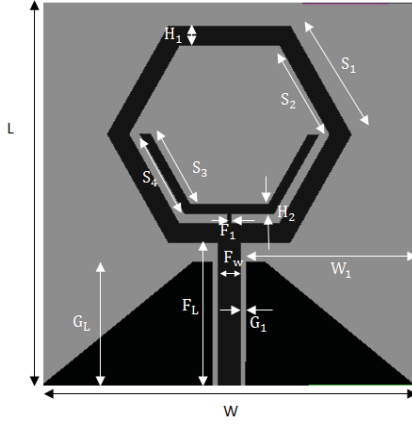


Fig.2: Dimensions of proposed antenna

$$f_{resonant} = \frac{ck_{mn}}{2\pi a_{eff}\sqrt{\epsilon_{eff}}} \quad (1)$$

$$a_{eff} = \frac{S_1 + S_2}{2} \quad (2)$$

When $\left(\frac{F_W}{X} < 1\right)$:

$$\epsilon_{eff} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \left[\left(1 + 12 \left(\frac{X}{F_W} \right)^{-1/2} \right) + 0.04 \left(1 - \left(\frac{F_W}{X} \right) \right)^2 \right] \quad (3)$$

When $\left(\frac{F_W}{X} > 1\right)$:

$$\epsilon_{eff} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \left(1 + 12 \left(\frac{X}{F_W} \right)^{-1/2} \right) \quad (4)$$

TABLE 1: Dimensions and parameters of proposed antenna

Parameters	Dimensions (mm)
L	20
W	20
S ₁	6.5
S ₂	5.3
S ₃	4.2
S ₄	4.76
H ₁	1
H ₂	0.5
F ₁	0.2
F _W	1.2
G ₁	0.3
G _L	6.5
F _L	7.5
W ₁	9.1

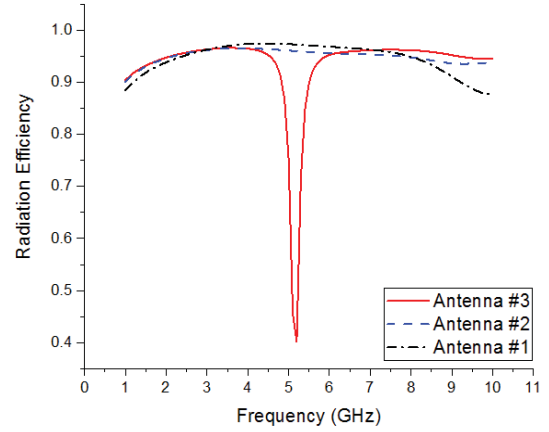


Fig.3: Simulated radiation efficiency of antenna design steps compared

III. RESULTS AND DISCUSSION

The working and parametric analysis is performed using High Frequency Structural Simulator (HFSS), which uses Finite Element Method (FEM) technique to validate the results.

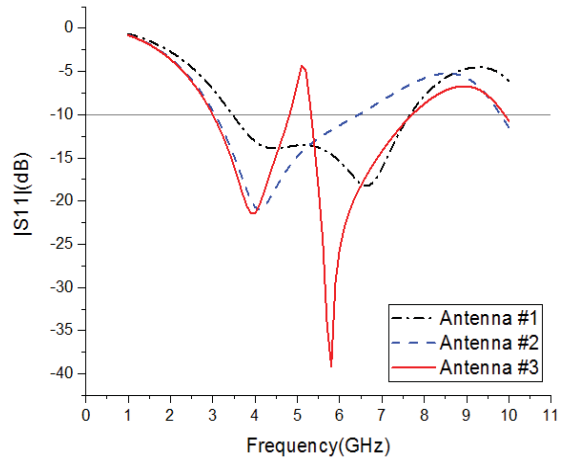


Fig.4: Comparative simulated S11 graphs of the antenna design steps

It can be observed that each stage has constant overall dimensions ($L \times W$). Fig.3 depicts the radiation efficiency of the three different antennas. Antenna (#1) has resonant frequencies at 4.4 GHz and 6.6 GHz. A broad bandwidth of 4.1 GHz (3.5 GHz – 7.6 GHz) is observed but lower frequencies are not covered. To achieve resonance at the lower side, the ground plane is tapered in antenna (#1), hence antenna (#2) is obtained as shown in Fig.1b. Antenna (#2) shows resonance at 4.1 GHz covering a bandwidth of 3.4 GHz (3 GHz – to 6.4Hz). Due to tapering of ground plane lower resonance is maintained but resonance at higher frequency is lost in antenna (#2). Tapering off the ground plane transforms the impedance linearly at the lower band. In square shaped ground plane resonance was obtained at 3.5GHz. After tapering the resonance shifted to 3GHz due to increased electrical length, it is depicted in antenna (#2). In order to compensate the loss of resonance at higher

frequency optimizations were done. The impedance versus frequency graphs were studied and it was observed that impedance was increasing due to stub loading so further tapering is carried out to reduce impedance and to maintain resonance at higher frequency. Antenna (#3) is derived from antenna (#2) structure by introducing a Y shaped stub onto the annular hexagonal ring keeping the overall dimensions unchanged as shown in Fig.1c.

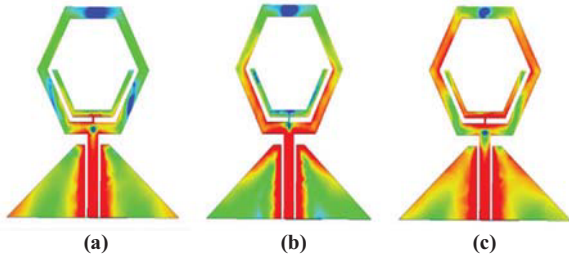


Fig.5
Simulated current distributions at (a) 3.5 GHz (b) 5.8 GHz (c) 7.4 GHz

Antenna (#3) performs well at both higher and lower frequency region. Creating notches at 3.9 GHz and 5.8 GHz antenna (#3) covers a bandwidth of 1.8 GHz (3 GHz – 4.8 GHz) and 2.4 GHz (5.3 GHz – 7.7 GHz) leading to a dual-band characteristics. Comparisons based on the return loss (S11) have been depicted in Fig.4 of the different evolution stages of the antenna. Fig.5 shows the current distributions at different frequencies of antenna (#3).

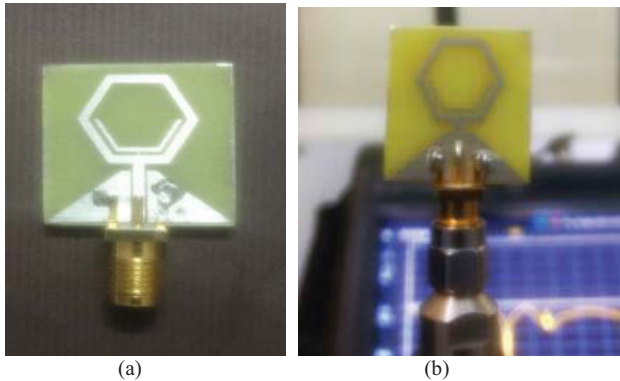


Fig.6: (a) Photograph of proposed antenna with SMA connector (b) Proposed antenna connected to VNA Tester

Observations were made that the annular ring radiated at lower frequencies while resonance at higher frequencies was obtained from the stub loading. The proposed stub loaded antenna parameters have been established based on parametric analysis. Measurement setups have been shown in Fig.6 to determine the bandwidth of the antenna using Vector Network Analyzer (VNA) and anechoic chamber.

TABLE 2: Comparative study if different antennas and proposed antenna

Reference	Antenna Size (mm ²)	Operating Frequencies Range (GHz)
5	30×30	2.9 - 7.9
6	40×40	5.02 - 10.84
7	28×26.5	4.75 - 11.1
10	50×50	1.48 - 1.87, 2.39 - 2.71, 3.02 - 3.16
Proposed Antenna	20×20	3 - 4.8, 5.3 - 7.7

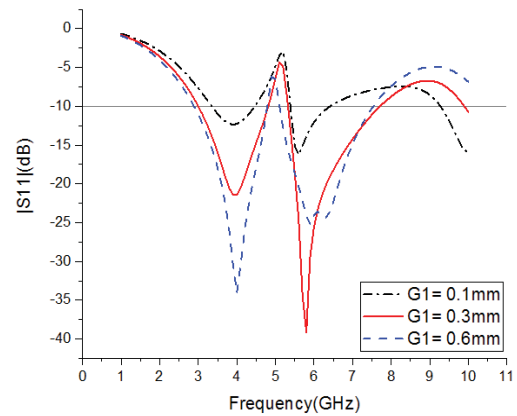


Fig.7: Parametric analysis of S11 by varying gap between signal and ground plane

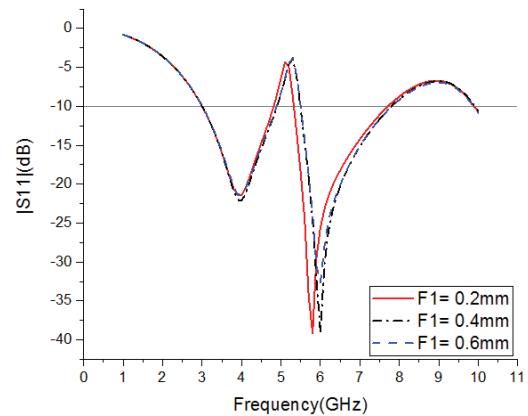


Fig.8: Parametric analysis of S11 by varying gap F1

Among the diverse parameters to be varied, the feed-ground plane gap (G1) and thickness of stub feed (F1) had major impact on the resonant frequencies. Fig.7 and Fig.8 depicts the shift in resonance for different values of the parameters. Hence for better coupling and wide bandwidth the parameters were optimized and selected. Fig.10 shows the voltage standing wave ratio (VSWR) of the proposed stub loaded antenna. Impedance matching is represented through this parameter.

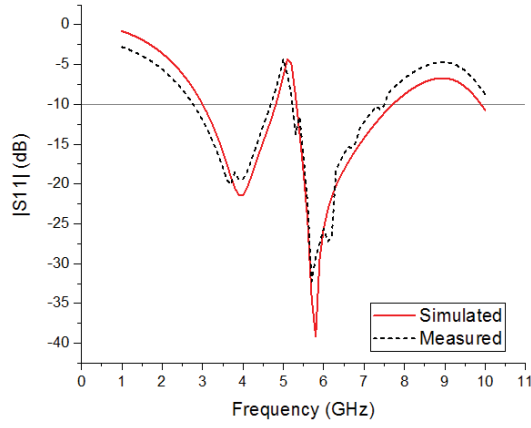


Fig.9: Measured and simulated S11 graphs of proposed antenna

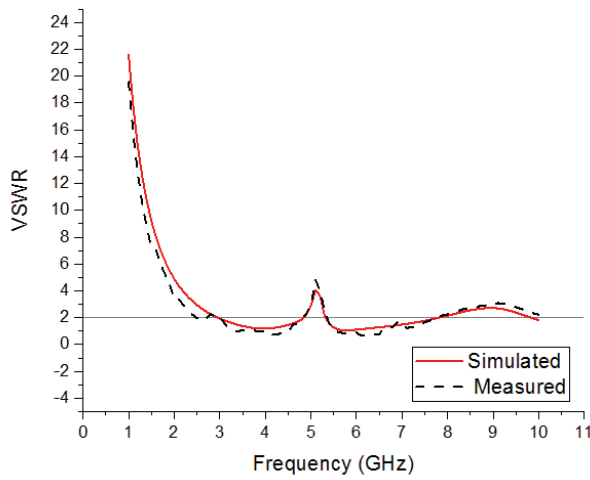


Fig.10: Simulated and measured VSWR graphs of proposed antenna

A VSWR of well below 2 is obtained in both resonance frequency bands. Hence impedance matching is efficiently achieved for dual band antenna. The radiation patterns of the proposed stub loaded antenna are shown in Fig.11 at 3.5 GHz and 5.8 GHz. The E-plane and the H-plane radiation characteristics are depicted from Fig.11 (a-d) were plotted and it was noted that the E and H plane co-polarization was omnidirectional. It is also inferred that the beam-width of the proposed antenna is high nearly ($\sim 60^\circ$). The proposed antenna with the other wideband microstrip antennas in the literature [5-7, 10] is compared based on size and bandwidth and presented in Table 2. Based on the comparative analysis the proposed antenna stands out for its miniature property and the wide bandwidth, compared to other wideband antennas in the literature.

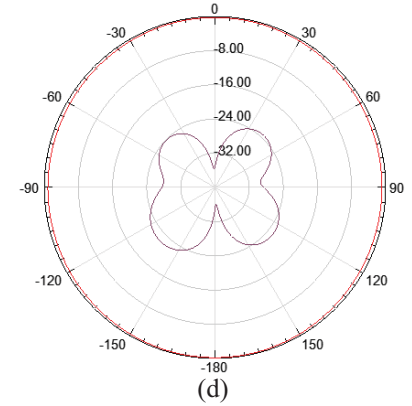
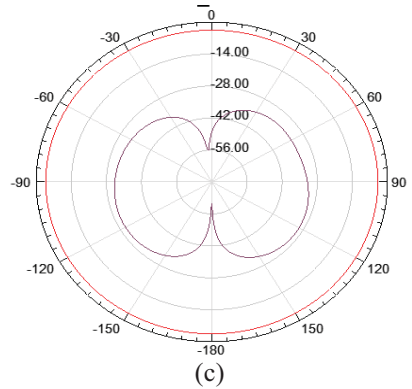
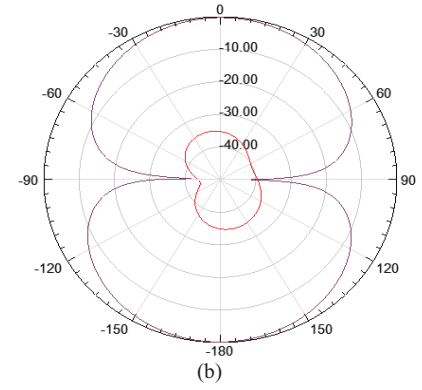
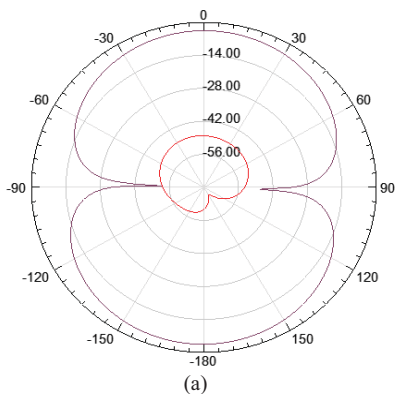


Fig.11: Simulated radiation patterns of the proposed antenna (a-b) 3.5 GHz and 5.8 GHz for xz -plane (c-d) 3.5 GHz and 5.8 GHz for yz -plane

IV. CONCLUSION

Compact stub loaded hexagonal ring antenna is designed and implemented for WLAN, WiMAX and ITU applications. The proposed antenna covers the entire band with broader bandwidth. Stable radiation pattern is achieved with low cross polarization. Experimental results such as return loss and VSWR are in well agreement with the simulated results of the proposed antenna. Miniaturization with multiband performance is achieved using ring type structure with stub loading and it makes the prototype a better candidate for wireless application. Gain of the prototype can be maximized using artificial magnetic conductor as ground plane.

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